

ACTIVITY 8: HACKERRANK ALGORITHMIC PROBLEM-SOLVING & PORTFOLIO INTEGRATION

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Semester: 3rd Semester

Portfolio links:

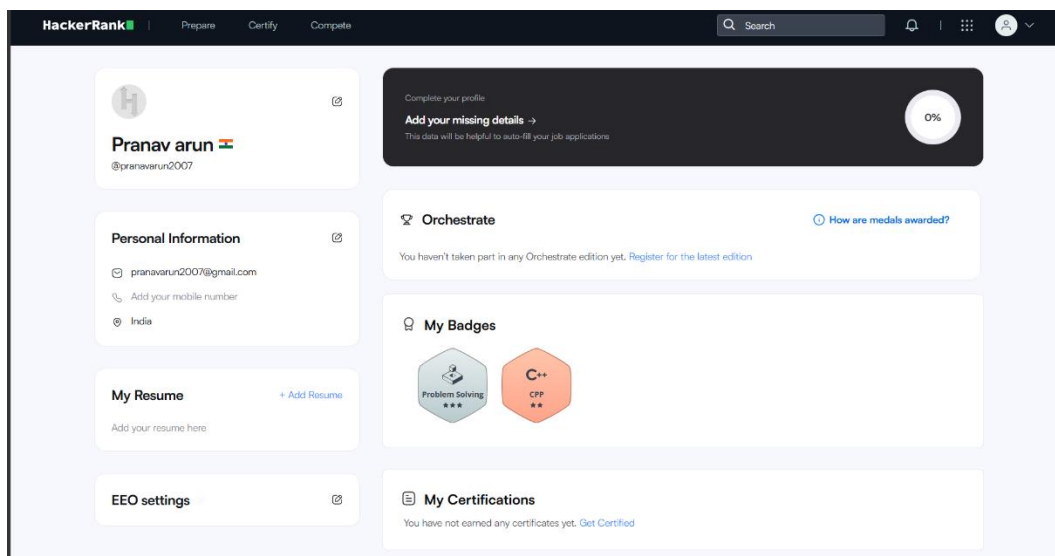
HackerRank Profile: <https://www.hackerrank.com/profile/pranavarun2007>

GitHub Repository: <https://github.com/Pranav-Arun18/HackerRank-3rdSem-Portfolio>

Problems completed:

Sl.no	Problem	Complexity analysis
1	Diagonal Difference	$O(N) / O(1)$
2	Dynamic Array	$O(N + Q) / O(N)$
3	Compare the Triplets	$O(1) / O(1)$
4	Time Conversion	$O(1) / O(1)$
5	Sparse Arrays	$O(N + Q) / O(N)$

HackerRank Badge:



Reflection

Solving the five HackerRank problems helped me improve my understanding of algorithmic problem-solving and efficient programming. The **Diagonal Difference** problem taught me how to work with a square matrix and efficiently calculate the sums of its primary and secondary diagonals. The **Dynamic Array** problem helped me understand how to handle multiple queries and work with dynamically changing arrays while keeping track of required values.

The **Time Conversion** problem improved my understanding of string manipulation and converting a time from 12-hour format to 24-hour format by handling cases such as AM and PM. **Compare the Triplets** showed me how a simple comparison-based approach can solve a problem efficiently when the input size is fixed. The **Sparse Arrays** problem helped me understand frequency counting and how storing the occurrence of strings can make repeated comparisons more efficient.

Through these problems, I also learned the importance of analyzing **time and space complexity** using Big-O notation. I understood that a correct solution should not only produce the expected output but should also use resources efficiently. Overall, these exercises improved my logical thinking, coding practice, and ability to choose suitable approaches for different types of problems. They also helped me organize my solutions and document my work as part of my GitHub portfolio.